Assign a scribe, messenger, and bookworm for the worksheet.

Scribe: _____ **Messenger:** _____ **Bookworm:** _____

Your TA will lend your group a 3D printed model of the graph $z = f(x, y)$ of some real-valued function f.

Please do NOT draw on the 3D printed model and return it to your TA at the end of tutorial.

Orientation: Assume the real-valued function $f : D \to \mathbb{R}$ has domain $D = [0, 6] \times [0, 6]$. The highest corner of your 3D model corresponds to the point $(0, 0) \in D$ and the lowest corner corresponds to the point $(6, 0) \in D$.

T7.1 Examine the 3D model with your group.

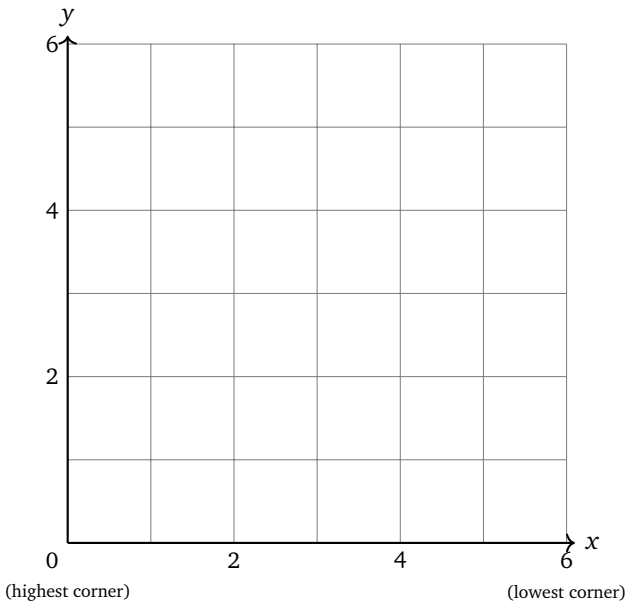
(a) Identify all of the critical points of f. Write their approximate coordinates.

(b) Classify the critical points that you found, e.g. "local max", "local min", "saddle point", "other".

(c) Identify all of the global extrema of f. Write their approximate coordinates.

(d) Identify all of the local extrema of f. Write their approximate coordinates.

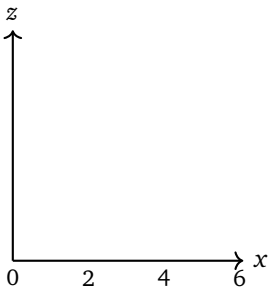
(e) On the grid below, draw and label all the points you found in the previous parts. Label them with their corresponding classifications, e.g. “local min”, “global max”, “saddle point”.



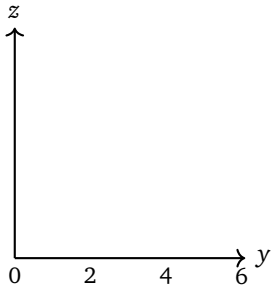
T7.2 Continue to use your 3D model according to the same domain $D = [0,6] \times [0,6]$ of the previous question.

(a) Let $a = (a_1, a_2)$ correspond to the local maximum of f lying inside the interior D^o.
Sketch the x-slice and y-slice of the graph of f passing through a.

y-slice at $y =$

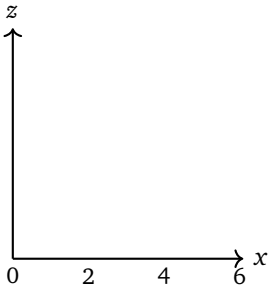


x-slice at $x =$

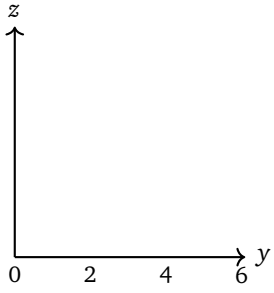


(b) Let $b = (b_1, b_2)$ correspond to the saddle point of f lying inside the interior D^o.
Sketch the x-slice and y-slice of the graph of f passing through b.

y-slice at $y =$

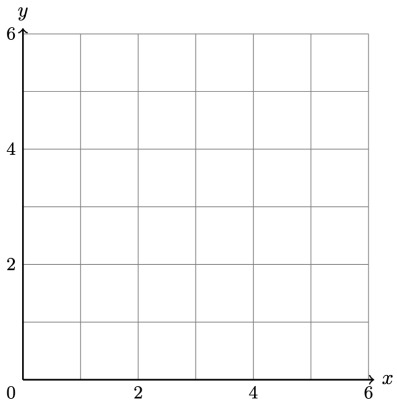


x-slice at $x =$

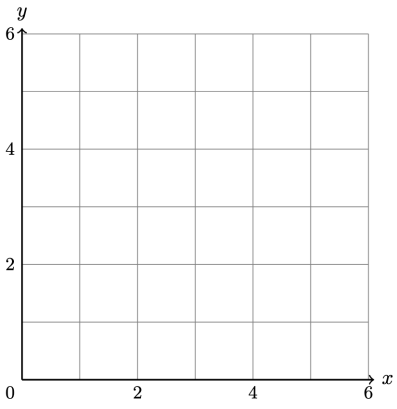


(c) Sketch the contour passing through a and the contour passing through b. Label a and b on each contour.

$z = f(a)$ contour



$z = f(b)$ contour



(d) After some time, your TA will share a link to a Math3D demo of the same graph. Play with the toggles and sliders to verify your group's answers to (a,b,c) above. Do they match? If not, what did you change?

T7.3 Your TA will share the function formula defining $f : D \to \mathbb{R}$ with $D = [0,6] \times [0,6]$. Record this formula below.

$$f(x,y) = _____$$

(a) Find the global maximum and minimum of f on D. Remember to justify your computations. You may use WolframAlpha to do basic algebraic manipulations and solve systems of polynomial equations; indicate when you have done so. *Hint:* Divide-and-conquer the domain with your group members.

(b) By the end of your calculations in (a), you must have checked a big list of candidate points for the global extrema of f on D. Does your list contain the same, fewer, or more points than what you found in T7.1? Explain why this might be.